Informatica® Cloud Data Integration

Oracle Autonomous Database Connector

Informatica Cloud Data Integration Oracle Autonomous Database Connector
October 2026

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Table of Contents

Preface 4

Informatica Resources 4

- Informatica Network 4
- Informatica Knowledge Base 4
- Informatica Documentation 4
- Informatica Product Availability Matrices 5
- Informatica Marketplace 5
- Informatica Global Customer Support 5

Chapter 1: Introduction to Oracle Autonomous Database Connector 6

Oracle Autonomous Database Connector assets 6

Chapter 2: Oracle Autonomous Database connections 7

Prerequisites 7

- Memory requirements 7
- Prepare for object storage authentication 7

Connect to Oracle Autonomous Database 8

- Before you begin 8
- Connection details 8
- Authentication types 9
- Object storage authentication types 10

Chapter 3: Mappings for Oracle Autonomous Database 11

Source properties for Oracle Autonomous Database 11

- Rules and guidelines for adding multiple source objects 13

Target properties for Oracle Autonomous Database 13

- Rules and guidelines for configuring override update query 15

Lookup transformation 16

- Lookup properties for Oracle Autonomous Database 17

Optimize the staging performance of a mapping 18

Rules and guidelines for mapping and mapping tasks 19

Appendix A: Data type reference 21

Oracle Autonomous Database data types and transformation data types 21

Rules and guidelines for data types 22

Preface

Use *Oracle Autonomous Database Connector* to learn how to read from or write to Oracle Autonomous Database by using Cloud Data Integration. Learn to create a Oracle Autonomous Database connection, run mappings and tasks in Cloud Data Integration.

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CHAPTER 1

Introduction to Oracle Autonomous Database Connector

You can use Oracle Autonomous Database Connector to read from and write to Oracle Autonomous Database instances hosted on Oracle Cloud Infrastructure.

Oracle Autonomous Database is an autonomous data management service in the cloud that performs routine database maintenance tasks, such as patching, upgrades, and tuning, while the system remains operational, all without human intervention. It provides an easy-to-use, fully autonomous database that scales elastically and delivers fast query performance. You can scale up or down the compute and storage capacity at any time without requiring downtime or service interruptions.

You can use an Oracle Autonomous Database connection as a source and as a target in mappings. You can create a mapping task to process data based on the data flow logic defined in a mapping or integration template.

Oracle Autonomous Database Connector assets

Create assets in Data Integration to integrate data using Oracle Autonomous Database Connector.

When you use Oracle Autonomous Database Connector, you can include the following Data Integration assets:

- Mapping
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

CHAPTER 2

Oracle Autonomous Database connections

Create an Oracle Autonomous Database connection to read data from or write data to Oracle Autonomous Database hosted on Oracle Cloud Infrastructure. You can use Oracle Autonomous Database connections to specify sources and targets in mappings and mapping tasks.

Prerequisites

Before you create an Oracle Autonomous Database connection to read from or write to Oracle Autonomous Database, be sure to complete the prerequisites.

Memory requirements

You can configure `INFA_MEMORY` and JVM options for the Secure Agent to optimize performance and avoid Java heap and memory-related errors when working with Oracle Autonomous Database Connector. Define these properties on the agent details page in Administrator.

For more information about the configurations required for the Secure Agent and detailed steps, see *INFA_MEMORY and JVM options* in the *Data Integration Performance Tuning* documentation.

Prepare for object storage authentication

You can configure the following object storage authentication methods for Oracle Autonomous Database Connector:

ConfigFile authentication

The ConfigFile authentication uses identity credentials of Oracle Cloud Infrastructure (OCI) account provided through a configuration file for authentication. This authentication method is based on the profile selected in the configuration file.

You can create a configuration file in the following format:

```
[<profile name>]
user=<user ocid>
fingerprint=<fingerprint>
tenancy=<tenancy ocid>
region=<region>
key_file=<private key file location>
```

You require the user OCID, fingerprint, and tenancy OCID information from the OCI account for the configuration file.

For more information about the steps to extract the identity credentials from the Oracle Cloud Infrastructure Console, see the Oracle Cloud Infrastructure documentation.

By default, the OCI configuration file is located at `~/.oci/config` on the Secure Agent machine. The `~/.oci/config` file can contain several profiles. The default profile name is `DEFAULT`. You can change the default profile name to any new profile names based on the profiles that you add to the `~/.oci/config` file. The `~/.oci/config` file cannot contain two profiles with the same name.

Simple authentication

The simple authentication uses API keys for authentication. You can provide the authentication details in the Oracle Autonomous Database connection. You need to place the private key file in the Secure Agent machine.

You require the user OCID, fingerprint, and tenancy OCID information from the Oracle Cloud Infrastructure account to create an Oracle Autonomous Database connection.

For more information about the steps to extract the identity credentials from the Oracle Cloud Infrastructure Console, see the Oracle Cloud Infrastructure documentation.

Connect to Oracle Autonomous Database

Let's configure the Oracle Autonomous Database connection properties to connect to Oracle Autonomous Database hosted on Oracle Cloud Infrastructure.

Before you begin

Before you get started, complete the following prerequisites:

- Configure the memory requirements for the connector.
- Get the required information from your Oracle Cloud Infrastructure account based on the object storage authentication type that you want to configure.

Check out "Prerequisites" on page 7 to learn more about this task.

Connection details

When you create an Oracle Autonomous Database connection, configure the connection properties.

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
User Name	The user name to access the Oracle Autonomous Database. It is used to authenticate the connection.
Password	The password associated with the user name.

Authentication types

You can configure TLS or mutual TLS authentication to connect to Oracle Autonomous Database. Select the required authentication type and then configure the authentication-specific parameters.
Default is mutual TLS authentication.

Mutual TLS authentication

The following table describes the basic connection properties for mutual TLS authentication:

Property	Description
Service Name	The specific database service accessed for the secured connection.
Wallet Path	The location of the wallet file for the secured connection.

TLS authentication

The following table describes the basic connection properties for TLS authentication:

Property	Description
TNS Name	The TNS name defined in the `tnsnames.ora` file. For example, `String tnsEntry = "(description= ... (security=(ssl_server_dn_match=yes)))"`

Object storage authentication types

You can configure ConfigFile or simple authentication for staging the connection when you connect to Oracle Autonomous Database. Select the required authentication type and then configure the authentication-specific parameters.
Default is ConfigFile authentication.

ConfigFile authentication

The following table describes the basic connection property for ConfigFile authentication:

Property	Description
Region	The Oracle Cloud Infrastructure region where the object storage bucket resides. Select the Oracle Cloud Object Storage region from the list.

Advanced settings

The following table describes the advanced connection properties for ConfigFile authentication:

Property	Description
Configuration File Location	The absolute path of the configuration file on the Secure Agent machine. If you do not enter the value, the Secure Agent uses the following configuration file path: `~/.oci/config`
Profile Name	The name of the profile in the configuration file that you want to use. Default is `DEFAULT`.

Simple authentication

The following table describes the basic connection properties for simple authentication:

Property	Description
User OCID	The unique identifier of the user in Oracle Cloud Infrastructure. For example, `ocid1.user.oc1..aaaaaaaaherdgpjknqzrwbdc7n5ksokkot7c5jngtx3pgolr7oqbw7xzksza`
Fingerprint	The fingerprint of the public key.
Tenancy OCID	The unique identifier of the tenancy in Oracle Cloud Infrastructure. The tenancy is the globally unique name of the Oracle Cloud Infrastructure account. For example, `ocid1.tenancy.oc1..aaaaaaaaba3pv6wkcr4jqae5f44n2b2m2yt2j6rx32uzr4h25vqstifsfdsq`
Private Key File Location	The location of the private key file in .PEM format on the Secure Agent machine.
Region	The Oracle Cloud Infrastructure region where the object storage bucket resides. Select the Oracle Cloud Object Storage region from the list.

CHAPTER 3

Mappings for Oracle Autonomous Database

You can configure a mapping to read from and write to Oracle Autonomous Database.

Use the Data Integration Mapping Designer to define and configure a mapping or mapping task for Oracle Autonomous Database sources and targets.

Source properties for Oracle Autonomous Database

In a mapping, you can configure a Source transformation to represent a Oracle Autonomous Database source.

The following table describes the Oracle Autonomous Database source properties that you can configure in a Source transformation:

Property	Description
Connection	Name of the source connection. You can select an existing connection, create a new connection, or define parameter values for the source connection property.
Source Type	Type of the source object. Select Single Object, Multiple Objects, Query, or Parameter. You can use implicit joins and advanced relationships with multiple objects. When you select the source type as query, you need to map all the fields selected in the query in the **Field Mapping** tab.
Object	The source object for the task. Select the source object for a single source. When you select the multiple source option, you can add multiple source objects and configure a relationship between them.

Property	Description
Parameter	A parameter file where you define values that you want to update without having to edit the task. The **Parameter** property appears only if you select parameter as the source type. Select an existing parameter for the source object, or click **New Parameter** to define a new parameter for the source object. If you want to overwrite the parameter at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the advanced session properties.
Filter	Filters records based on the filter condition. You can specify a simple filter or an advanced filter.
Sort	Sorts records based on the conditions you specify. You can specify the following sort conditions: - Not parameterized. Select the fields and type of sorting to use. - Parameterized. Use a parameter to specify the sort option.
Select distinct rows only	Extracts distinct rows from the source table. **Note:** If you select the source type as query or you specify the SQL query property, the **Select distinct rows only** option is ignored.

The following table describes the advanced properties that you can configure in a Source transformation:

Property	Description
Bucket Name	The Oracle Cloud Object Storage bucket name that contains the objects.
Staging Directory Path	Path of the local staging directory. Ensure that the user has write permissions on the directory. In addition, ensure that there is sufficient space to enable staging of the entire file. Default staging directory is the `/temp` directory on the machine that hosts the Secure Agent. The temporary files are created within the new directory.
EXPORT_DATA Command Options	The options to provide in the EXPORT_DATA command when loading the data from Oracle Autonomous Database to Oracle Cloud Object Storage. For example, `DELIMITER=|`
Schema Name	Overrides the schema name of the source object.
Table Name	Overrides the table name configured in the source object and uses the specified table name to read data at runtime.
Pre-SQL	The pre-SQL command to run on the Oracle Autonomous Database source table before the agent reads the data. For example, if you want to update records in the database before you read the records from the table, specify a pre-SQL statement. The query must include a fully qualified table name.
Post-SQL	The post-SQL command to run on the Oracle Autonomous Database table after the agent completes the read operation. For example, if you want to delete some records after the latest records are loaded, specify a post-SQL statement. The query must include a fully qualified table name.

Property	Description
SQL Override	The SQL statement to override the default query used to read data from the Oracle Autonomous Database source.
Tracing Level	Determines the amount of detail that appears in the log file. You can select Terse, Normal, Verbose Initialization, or Verbose Data. Default is Normal.

Rules and guidelines for adding multiple source objects

Consider the following rules and guidelines when you add multiple source objects:

- When you configure an implicit join, ensure that all tables involved in the join are available in the same schema.
- When you configure a mapping, ensure that source objects from multiple sources do not contain repeated columns with the same name. If tables have columns with the same name, you cannot map them to the target.
  For example, in `CustomerTable(Id, FirstName, LastName)` and `OrderTable(Id, Amount)`, you cannot connect the duplicate `Id` column to the target.
- You cannot use a self join when adding multiple source tables.
- When you specify a cross join condition in an advanced relationship to read data from multiple tables, use the following format:
  `SchemaName1.TableName1 JOIN SchemaName2.TableName2 ON SchemaName1.TableName1.Column1 = SchemaName2.TableName2.Column2`
- When you configure an advanced relationship to join multiple data sources, if a column name includes special characters, you need to enclose the column name in quotes.

Target properties for Oracle Autonomous Database

In a mapping, you can configure a Target transformation to represent a Oracle Autonomous Database target.

The following table describes the Oracle Autonomous Database target properties that you can configure in a Target transformation:

Property	Description
Connection	Name of the target connection. You can select an existing connection, create a new connection, or define parameter values for the target connection property.
Target Type	Type of the target object. Select **Single Object** or **Parameter**.

Property	Description
Object	Name of the target object. You can select an existing object. The target name can contain alphanumeric characters. You can use only a period (.), an underscore (_), an at the rate sign (@), a dollar sign ($), and a percentage sign (%) special characters in the file name. You can use parameters defined in a parameter file in the target name.
Parameter	A parameter file where you define values that you want to update without having to edit the task. The **Parameter** property appears only if you select parameter as the target type. Select an existing parameter for the target object, or click **New Parameter** to define a new parameter for the target object. If you want to overwrite the parameter at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the advanced session properties.
Operation	The target operation. Select one of the following operations: - Insert - Update - Upsert - Delete - Data Driven Select **Data Driven** if you want to create a mapping to capture changed data from a CDC source.
Data Driven Condition	Appears only when you select **Data Driven** as the operation type. Defines expressions to flag rows for an insert, update, or delete operation. Specify the data driven condition for non-CDC sources. For CDC sources, leave the field empty as the rows in the CDC source tables are already marked with the operation types.
Update Columns	The primary key column to update, upsert, or delete data in an Oracle Autonomous Database target. This field is not required if the target table already has a primary key. If you do not specify a primary key column, the mapping considers the target column that you configured as the primary key column to update, upsert, or delete data.
Create Target	Creates a new target. When you create a new target, enter a value of the following fields: - **Name**: Enter a name for the target object. - **Path**: Name of an existing schema to store the target table. By default, the field is empty.

The following table describes the advanced properties that you can configure in a Target transformation:

Note: You can parameterize all the target advanced properties.

Property	Description
Bucket Name	The Oracle Cloud Object Storage bucket name that contains the objects.
Local Staging Directory Path	Path of the local staging directory to store temporary files. Ensure that the user has write permissions on the directory. In addition, ensure that there is sufficient space to enable staging of the entire file. Default staging directory is the `/temp` directory on the machine that hosts the Secure Agent.

Property	Description
Truncate Table	Truncates the target table before loading data to the target. Default is disabled.
COPY_DATA Command Options	The options to provide in the COPY_DATA command when loading the data from Oracle Cloud Object Storage to Oracle Autonomous Database. For example, `ignoremissingcolumns : true`
Schema Name	Overrides the schema name of the target object.
Table Name	Overrides the table name configured in the target object and uses the specified table name to write data at runtime.
Pre-SQL	SQL statement that you want to run before writing data to the target. For example, if you want to delete specific records from the database before you write the records into the table, specify the following pre-SQL statement: `delete from <schema_name>.<table_name> where id between 5 and 9;`
Post-SQL	SQL statement that you want to run after writing the data into the target. For example, if you want to delete specific records in a table after you write the records into the target table, specify the following post-SQL statement: `delete from <schema_name>.<table_name> where id between 5 and 9;`
Override Update Query	Overrides the default update query that the Secure Agent generates for the update operation with the update query that you specify. For more information about configuring override update query, see "Rules and guidelines for configuring override update query" on page 15.
Batch Size	The number of rows to load in a batch. The default value is 2,000,000.
Number of files per batch	The number of files in a batch. The default value is 20.
Forward Rejected Rows	Determines whether the transformation passes rejected rows to the next transformation or drops rejected rows. By default, the agent forwards rejected rows to the next transformation.

Rules and guidelines for configuring override update query

You can specify a update query override to override the update query that the Secure Agent generates for the update operation.

Consider the following rules and guidelines when you use the override update query property for an Oracle Autonomous Database target:

- You cannot use the override update query for an upsert operation.
- You cannot specify multiple override update queries for an update operation.
- Specify the override update query in the following format:

```
UPDATE <Target_Schema_Name>.<Target_Table_Name>
  SET (<update_column1>,<update_column2>) = (
    SELECT {{TU}}.<update_column1>, {{TU}}.<update_column2>
    FROM {{TU}}
    WHERE <Target_Schema_Name>.<Target_Table_Name>.<update_column3>=
{{TU}}.<update_column3>
```

```
    )
  WHERE EXISTS (
    SELECT 1
    FROM {{TU}}
    WHERE <Target_Schema_Name>.<Target_Table_Name>.<update_column1> =
{{TU}}.<update_column1>
    AND <Target_Schema_Name>.<Target_Table_Name>.<update_column2> =
{{TU}}.<update_column2>
    AND <Target_Schema_Name>.<Target_Table_Name>.<update_column3> =
{{TU}}.<update_column3>
    );
```

Example:

```
UPDATE ADMIN.global_normal_employees
SET (EMP_ID, SALARY) = (
    SELECT {{TU}}.EMP_ID, {{TU}}.SALARY
    FROM {{TU}}
    WHERE ADMIN.global_normal_employees.EMP_NAME = {{TU}}.EMP_NAME
)
WHERE EXISTS (
    SELECT 1
    FROM {{TU}}
    WHERE ADMIN.global_normal_employees.EMP_NAME = {{TU}}.EMP_NAME
)
```

- The column names for :TU must match the target table column names.
- The `WHERE` column is mandatory in the query.
- You cannot use left, right, or full outer joins in the `FROM` column of an override update query statement.
- All the column names in the query must be qualified names and the table name and schema name must be associated with the column names.
- You must specify the override update query with a valid SQL syntax because the connector replaces :TU with a temporary table name and does not validate the update query.
- You cannot change the order of the column mappings using the override update query.

Lookup transformation

You can create lookups for objects in an Oracle Autonomous Database mapping. You can retrieve data from an Oracle Autonomous Database lookup object based on the specified lookup condition.

Use the Lookup transformation to look up data in an Oracle Autonomous Database object. For example, the source table includes the customer code, but you want to include the customer name in the target table to make summary data easy to read. You can use the Oracle Autonomous Database Lookup transformation to look up the customer name in another Oracle Autonomous Database object.

You can add the following lookups to a mapping:

- Connected with cached
- Connected with uncached
- Unconnected with cached
- Dynamic lookup cache

For more information about the Lookup transformation, see *Transformations* in the Data Integration documentation.

Lookup properties for Oracle Autonomous Database

The following table describes the Oracle Autonomous Database lookup object properties that you can configure in a Lookup transformation:

Property	Description
Connection	Name of the lookup connection. You can select an existing connection, create a new connection, or define parameter values for the lookup connection property. If you want to override the lookup connection properties at runtime, select the **Allow parameter to be overridden at run time** option.
Source Type	Type of the source object. Select Single Object, Query, or Parameter. **Note:** You cannot configure uncached lookups when you select query as the source type.
Parameter	A parameter file where you define values that you want to update without having to edit the task. Select an existing parameter for the lookup object or click **New Parameter** to define a new parameter for the lookup object. The **Parameter** property appears only if you select parameter as the lookup type. When the task runs, the Secure Agent uses the parameters from the file that you specify in the advanced session properties.
Lookup Object	Name of the lookup object for the mapping.
Multiple Matches	Behavior when the lookup condition returns multiple matches. You can return all rows, any row, the first row, the last row, or an error. You can select from the following options in the lookup object properties to determine the behavior: - Return first row - Return last row - Return any row - Return all rows - Report error

The following table describes the Oracle Autonomous Database advanced lookup properties that you can configure in a Lookup transformation:

Property	Description
Bucket Name	The Oracle Cloud Object Storage bucket name that contains the objects.
Staging Directory Path	Path of the local staging directory. Ensure that the user has write permissions on the directory. In addition, ensure that there is sufficient space to enable staging of the entire file. Default staging directory is the `/temp` directory on the machine that hosts the Secure Agent. The temporary files are created within the new directory.
EXPORT_DATA Command Options	The options to provide in the EXPORT_DATA command when loading the data from Oracle Autonomous Database to Oracle Cloud Object Storage. Example: `DELIMITER=|`
Schema Name	Overrides the default schema name. **Note:** You cannot configure a custom query when you use the schema name.

Property	Description
Table Name	Overrides the default source table name. **Note:** When you select multiple objects or query as the source type, you cannot use this property.
Pre-SQL	The pre SQL command to run on the Oracle Autonomous Database source table before the agent reads the data. For example, if you want to update records in the database before you read the records from the table, specify a pre SQL statement. The query needs to include a fully qualified table name.
Post-SQL	The post SQL command to run on the Oracle Autonomous Database table after the agent completes the read operation. For example, if you want to delete some records after the latest records are loaded, specify a post SQL statement. The query needs to include a fully qualified table name.
SQL Override	The SQL statement to override the default query used to read data from the Oracle Autonomous Database source.
Tracing Level	Determines the amount of detail that appears in the log file. You can select Terse, Normal, Verbose Initialization, or Verbose Data. Default is Normal.
Lookup Data Filter	Limits the number of lookups that the mapping performs on the cache of the lookup source table based on the value you specify in the filter condition. Maximum length is 32768 characters.

For more information about the Lookup transformation, see *Transformations* in the Data Integration documentation.

Optimize the staging performance of a mapping

When you configure a mapping, you have the option to read or look up data from the Oracle Autonomous Database source with or without staging data on Oracle Cloud Object Storage. However, you must configure staging mode in the write operation.

You can configure the read mode as either direct or staging in the Source and Lookup transformations.

Staging read mode

In the staging read mode, the agent creates a staging file in the directory that you specify in the mapping properties. You need to specify a staging directory on the Secure Agent machine that has sufficient disk space corresponding to the volume of data that you want to process.

When you run the mapping, the `EXPORT_DATA` command loads the data from the Oracle Autonomous Database table to the Oracle Cloud Object Storage bucket, and then the agent loads the data from Object Storage bucket to the staging directory on the agent machine. After the agent completes the read operation, the staging files are deleted.

Direct read mode

In the direct read mode, Data Integration directly accesses and retrieves the data from Oracle Autonomous Database without staging the data.

The advanced source attributes applicable for Oracle Cloud Object Storage staging in the mapping properties and the staging optimization property in the agent properties are not required for the direct read mode.

Setting the staging property

You can optimize the staging performance of read, lookup, and write operations.

You need to set the following staging property in the agent properties based on the operation that you want to optimize:

- To optimize the read operation, set the following property: `INFA_DTM_RDR_STAGING_ENABLED_CONNECTORS`
- To optimize the lookup operation, set the following property: `INFA_DTM_LKP_STAGING_ENABLED_CONNECTORS`
- To optimize the write operation, set the following property: `INFA_DTM_STAGING_ENABLED_CONNECTORS`

Perform the following tasks to set the staging property for the Tomcat in the Secure Agent properties:

1. In Administrator, click **Runtime Environments**.
2. Edit the Secure Agent for which you want to set the property.
3. In the **System Configuration Details** section, select the **Service** as **Data Integration Server** and the type as **Tomcat**.
4. Set the value of the Tomcat property to the plugin ID of Oracle Autonomous Database Connector. You can find the plugin ID in the manifest file located in the following directory: `<Secure Agent installation directory>/downloads/<Oracle Autonomous Database package>/CCIManifest`

When you enable the staging mode, you can see the following message in the logs:

- Read operation: `READER_1_1_1> SDKS_38636 [2025-07-04 12:04:52.743] Plug-in #648300: DTM Staging is enabled for connector for Source Instance [Source].`
- Lookup operation: `LKPDP_1:READER_1_1> SDKS_38636 [2025-07-04 15:07:23.020] Plug-in #648300: DTM Staging is enabled for connector for Source Instance [Source].`
- Write operation: `INFA_DTM_STAGING mode is enabled for the write operation.`

Rules and guidelines for mapping and mapping tasks

Consider the following rules and guidelines when you configure mappings and mapping tasks:

Target transformation

When you create the target table at runtime in a mapping, fields with string data types are written as NVARCHAR2 data type in the target table. Additionally, the precision for these fields is limited to 16380. If you write data with a higher precision value, the mapping fails.

Lookup transformation

- By default, you cannot use null values in the filter condition for the connected Lookup transformation. To use null values in the filter condition, set the `ENABLE_NULL_FLAG_FOR_UNCACHED_LOOKUP=true` property as a JVM option under the DTM type in the Data Integration Server service on the Secure Agent.
- When you use a string column as the lookup field in a cached Lookup transformation and write to a flat file target, the mapping handles null values in the lookup condition differently depending on the configured read mode. If you use direct read mode, the mapping writes the null value in the string column as null in the target. However, if you use staging optimization, the mapping writes a null value in the string column as an empty string in the target.

APPENDIX A

Data type reference

Data Integration uses the following data types in mappings and mapping tasks with Oracle Autonomous Database:

Native data types

Oracle Autonomous Database data types appear in the Source and Target transformations when you choose to edit metadata for the fields.

Transformation data types

Set of data types that appear in the transformation. They are internal data types based on ANSI SQL-92 generic data types, which Data Integration uses to move data across platforms. Transformation data types appear in all transformations in a mapping, or mapping task.

When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

Oracle Autonomous Database data types and transformation data types

Oracle Autonomous Database data types map to transformation data types that Data Integration uses to move data across platforms.

The following table lists the Oracle Autonomous Database data types that you can use withOracle Autonomous Database Connector, along with the corresponding transformation data types:

Oracle Autonomous Database data type	Transformation data type	Range and description for the transformation data type
CHAR	String	Minimum size: 1 byte Maximum size: 2,000 bytes or characters Default size: 1 byte
DATE	Date/Time	Date range: January 1, 4712 BC to December 31, 9999 AD The data type contains the datetime fields `YEAR`, `MONTH`, `DAY`, `HOUR`, `MINUTE`, and `SECOND`. It does not include fractional seconds or time zone.

Oracle Autonomous Database data type	Transformation data type	Range and description for the transformation data type
NCHAR	String	Minimum size: 1 character Maximum size: 2,000 bytes Default size: 1 character
NUMBER	Decimal	Precision range: 1 to 38 Scale range: -84 to 127 Precision of 1 to 28 and scale of 0 to 28.
NVARCHAR2	String	Minimum size: 1 byte or character Maximum size: 32,767 bytes or characters
TIMESTAMP	Date/Time	Year, month, and day values of date and hour, minute, and second values of time. The `fractional_seconds_precision` provides the number of digits in the fractional part of the `SECOND` Date/Time field. You can enter a value between 0 to 9 in the `fractional_seconds_precision` field and the default value is 6.
TIMESTAMP WITH TIME ZONE	String	Size: 128 characters Contains date, time, and an explicit time zone offset.
TIMESTAMP WITH LOCAL TIME ZONE	String	Size: 128 characters Contains date and time normalized to the database time zone.
VARCHAR2	String	Minimum size: 1 byte or character Maximum size: 32,767 bytes or characters

Rules and guidelines for data types

You need to follow certain rules and guidelines for some of the data types.

Number

When you write to a Number data type, if the precision at the source exceeds 28, it is limited to 28 in the target, and the scale adjusts accordingly. If the scale becomes negative after adjustment, it is set to 0. If the scale exceeds the adjusted precision, it is also reduced to match the precision.

The following table provides examples of how source data with different precision and scale is transformed at the target:

Data at source	Precision and scale at source	Data at target	Precision and scale at target
1.2345678901234567890123456789012345678	38, 37	1.2345678901234567890123456789	28, 27
0.0000000000123456789012345678901234567 8	28, 38	0.0000000000123456789012345679	28, 28
1234567890123.456789012345678901234567 8	38, 25	1234567890123.456789012345679	28, 15

Timestamp

When you use a Filter transformation to filter date or timestamp data types, use the `TO_DATE()` expression in the following format:

```
TO_DATE('1970-01-01 23:05:06.123457', 'YYYY-MM-DD HH24:MI:SS.US')
```

Ensure that the source string format matches the format string, and adjust the fractional part of the string based on the source values.

For more information on `TO_DATE` format strings, see the *Transformation Language Reference Guide*.